Technical Data Sheet

NebuLink Module
NLN500b – LoRa Module

Document Date: November, 2023

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ds_nln500b_2.0

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Preface

Privacy Information

Table below shows the revision history of NebuLink Node NLN 500b – an industrial LoRa module data sheet.

Revision History

Version	Date	Description of Changes
1.0	September, 2023	Initial Release
2.0	November, 2023	Final Release

Typographic Conventions

This data sheet uses the typographic conventions as shown below:

Visual Cue	Meaning
Bold Type with Initial Capital letters	All headings and Sub headings Titles in a document are displayed in bold type with initial capital letters; Example: General Description, Features.
Bold Type with Italic Letters	All Definitions, Figure and Table Headings are displayed in Italics. Examples: ***Figure 1-1. NebuLink Node NLN 500b, Table 1-1. NebuLink Node NLN 500b Specifications.***
1., 2.	Numbered steps are used in a list of items, when the sequence of items is important. such as steps listed in procedure.
•	Bullets are used in a list of items when the sequence of items is not important.

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1 Introduction

The NebuLink Module NLN 500b is smallest STM32WL based LoRa Module. It is build on the STM32WL SoC which has Embedded LoRaWAN Transceiver for wide range of Smart Applications. It is useful when long range is required as it operates at 865/868 MHz band and uses LoRa (Long-Range) and Sigfox Protocol for communication. It offers long range, low power and secure data transmission for M2M and IoT applications. This Module designed to be extremely low-power and is based on the high-performance Arm® Cortex®-M4 32-bit RISC core operating at a frequency of up to 48 MHz. This core implements a full set of DSP instructions and an independent memory protection unit (MPU) that enhances the application security.

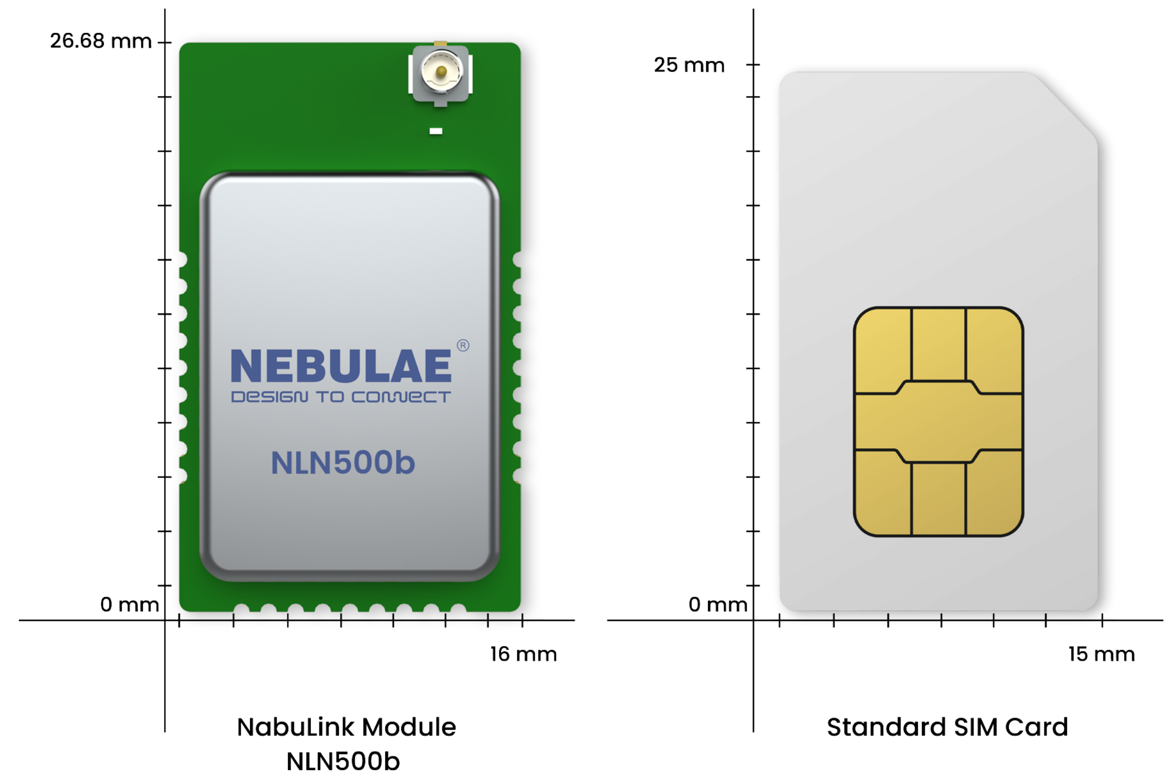

Figure 1: NLN 500b – LoRa based NebuLink Module

2 Key Features

- Frequency Band: Supports EU 863-870, IN865
- RX Sensitivity: –123 dBm (125 kHz BW and spreading factor 7) and –136 dBm (125kHz BW and spreading factor 12) for LoRa
- Transmitter Output Power: High output power programmable up to +21 dBm
- Supported Protocols: LoRaWAN® and Sigfox™
- Power Supply: 1.8V to 3.6 VDC
- Operating Temperature Range: -20°C to +75°C
- Core: 32-bit Arm®Cortex®-M4 CPU
- Hardware Encryption: AES 256-bit
- Memory: Up to 256-Kbyte Flash memory, Up to 64-Kbyte RAM
- Update: OTA (over-the-air) firmware update capable
- Programming/Debug: Serial-wire debug (SWD), JTAG
- I/Os: up to 21 I/Os
- I/O Tolerance Voltage: 5VDC
- The Module can be directly soldered on any device board
- RF shield on the module for preventing the module from external electromagnetic radiation and interference

3 Brief Specification

Table – 1 Module Size

Module	Chip	Power (dBm)	Antenna	Dimension (mm) LxWxH
NLN 500b	STM32WLE5CBU6	+21	External	26.68×16.0×3.6

4 Application

The applications include, but are not limited to:

- Building automation & monitoring
- Lighting controls
- Inventory management
- Environmental monitoring
- Security
- Industrial monitoring
- Machinery condition and performance monitoring
- Monitoring of plant system parameters such as temperature, pressure, flow, tank level, humidity, vibration, etc.
- Smart Metering & Lighting Solutions
- Smart Flood Sensors
- Smart Street Lighting
- Smart City Network
- Smart Waste Management

5 Ordering Information

Table – 2 Ordering Information

Part Number	RF Output
PL1N500BUF100	u.FL
PL1N500BFL100	Feedline
PL1N500SGL100	Spring Antenna

6 Block Diagram

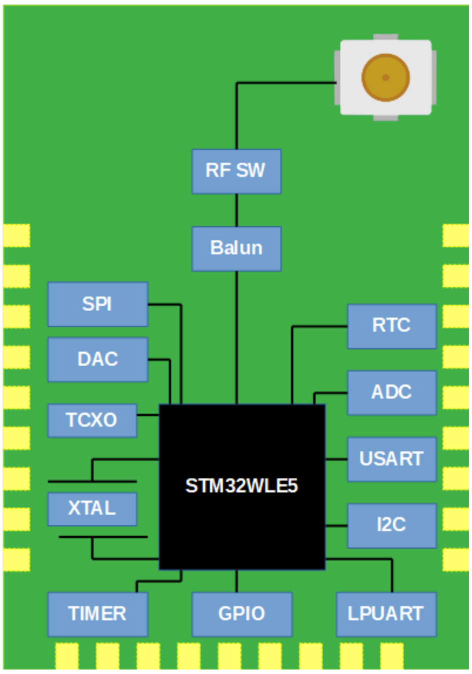

Figure 2: Block Diagram

7 Pin Details

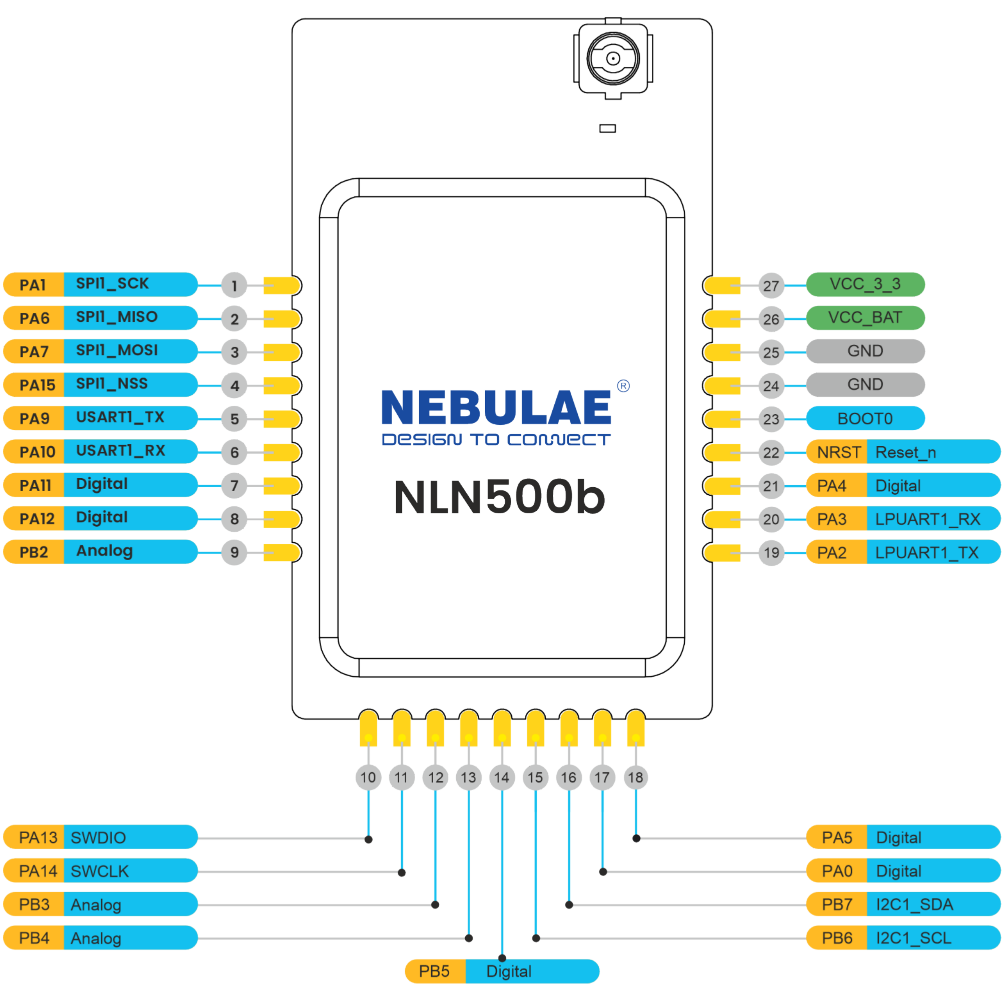

Figure 3: Pinout Diagram

Table – 3 Pin Details

Pin Number	Ports	Ports Description
1	PA1	SPI1_SCK
2	PA6	SPI1_MISO
3	PA7	SPI1_MOSI
4	PA15	SPI1_NSS
5	PA9	USART1_TX
6	PA10	USART1_RX
7	PA11	Digital
8	PA12	Digital
9	PB2	Analog

10	PA13	SWDIO
11	PA14	SWCLK
12	PB3	Analog
13	PB4	Analog
14	PB5	Digital
15	PB6	I2C1_SCL
16	PB7	I2C1_SDA
17	PA0	Digital
18	PA5	Digital
19	PA2	LPUART1_TX
20	PA3	LPUART1_RX
21	PA4	Digital
22	NRST	Reset_n
23	BOOT0	Boot
24	GND	GND
25	GND	GND
26	VCC_BAT	VCC_BAT
27	VCC_3_3	VCC_3_3

7.1 Peripherals

Table – 4 Peripherals

Interface	Quantity
I2C	1, programmable up to 3
SPI	1, programmable up to 2
U(S)ART	1
LPUART	1

Table below provides the information about the usage of the I/Os available in NLN 500b.

Table – 5 I/O Details

Usage	Quantity
Digital Pin	Up to 21 pins
ADC	8 pins
Comparators	2 pins
DAC	1 pin

8 Operating Characteristics

Table – 6 Operating Characteristics					
Parameter		**Minimum**	**Typical**	**Maximum**	**Unit**
Temperature		–20		75	°C
Supply Voltage (VDD)		1.8	3.3	3.6	V
Supply Voltage Ripple or Noise (VDD)				10	mV
Frequency	IN	865		867	MHz
	EU	863		870	MHz
TCXO Frequency			32		MHz
XTAL Frequency			32.768		KHz
XTAL Frequency Tolerance			±20ppm		
Power consumption(sleep mode)			1.7		mA
Power consumption(standby mode)			6.95		mA

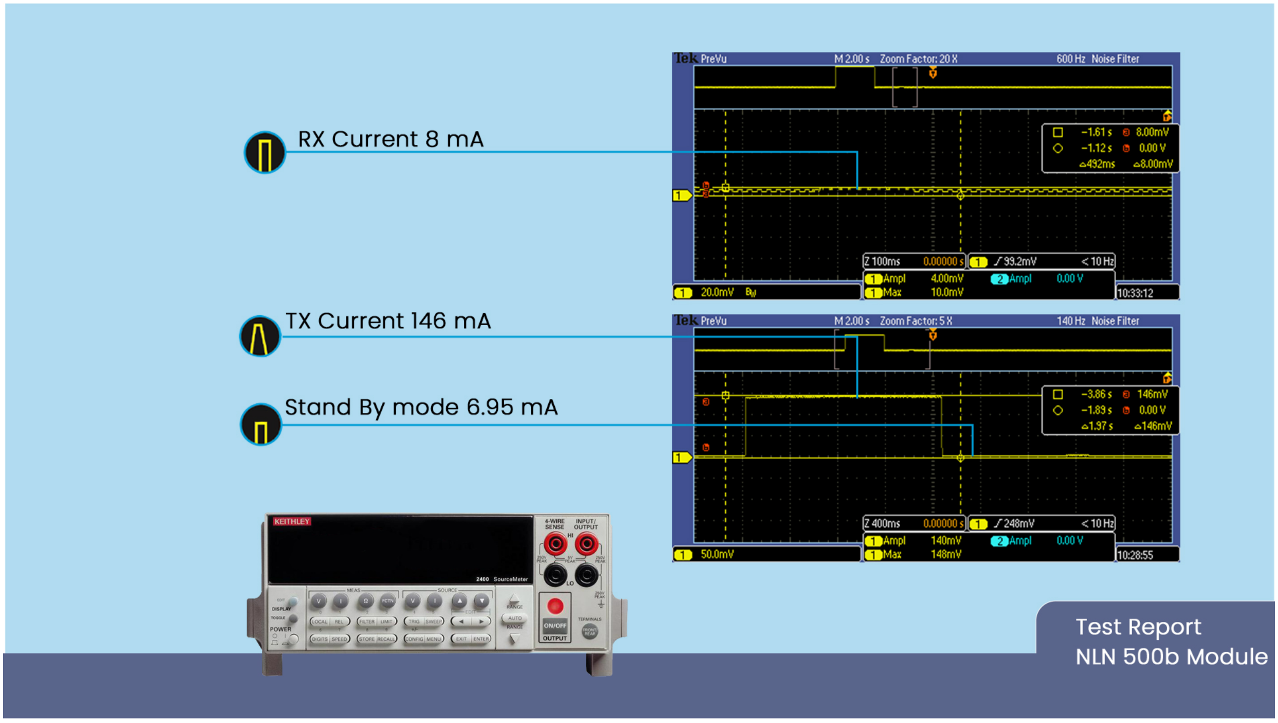

Figure 4: Operating Characteristic Test Report

9 RF Characteristic

Table – 8 RF Characteristic

Parameter	Minimum	Typical	Maximum	Unit
Output RF level (High Power Mode)		+21		dBm
Power consumption (PA=+21dBm), TX Mode	-	-	146	mA
Power consumption (PA=+21dBm), RXMode			8	mA
Power consumption (PA=+13dBm), TX Mode	-	-	106	mA
Power consumption (PA=+13dBm), RX Mode			8	mA
Sensitivity (865Mhz, BW=125Khz SF=12)		-136		dBm
Sensitivity (865Mhz, BW=125Khz SF=7)		-123		dBm
Frequency Band	IN865, EU 863-870			
Modulation	LoRa/(G)FSK			

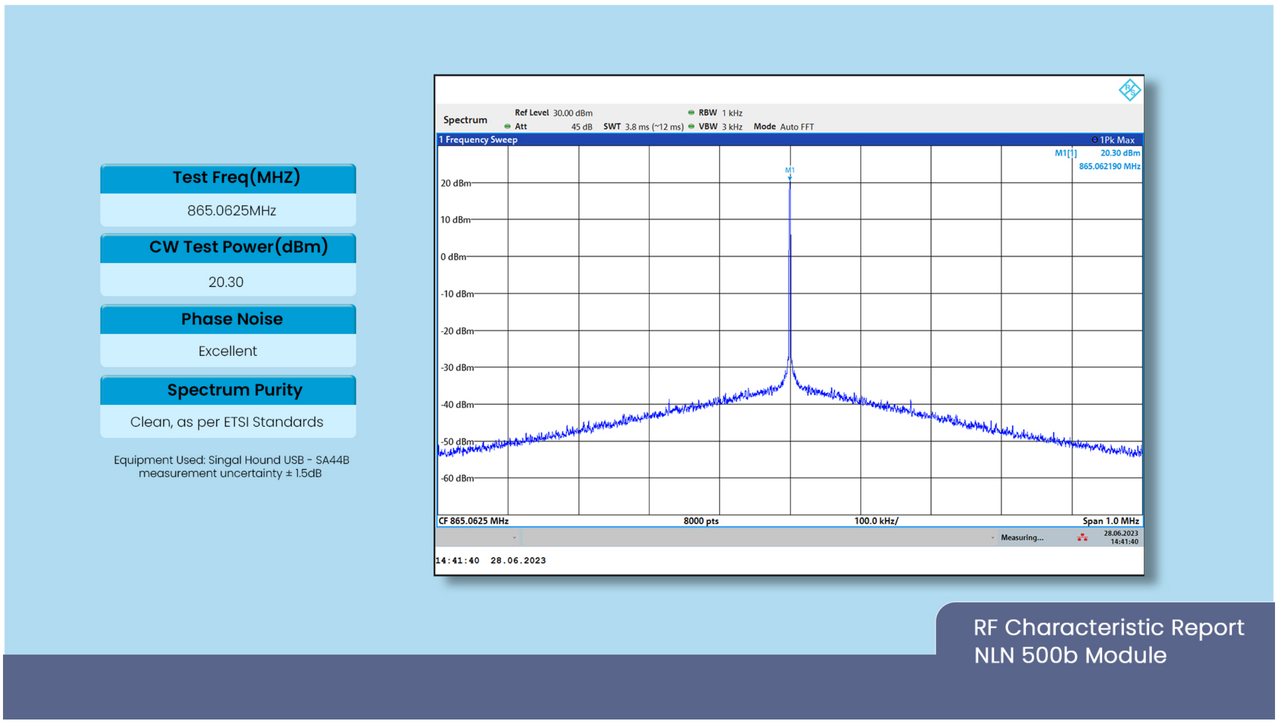

Figure 5: RF Characteristic Report

10 Program and Debug

1. To generate the code for STM32WLE5CBU6 microcontroller download the STM32CubeMX tool from below link:

   https://www.st.com/en/development-tools/stm32cubemx.html

2. To compile and debug the code, download STM32CubeIDE from below link,

   https://www.st.com/en/development-tools/stm32cubeide.html

3. To programme the binary, download STM32CubeProgrammer from below link,

   https://www.st.com/en/development-tools/stm32cubeprog.html

4. Use below pins to programme the NLN 500b module:
   1. GND
   2. 3V3
   3. JTMS-SWDIO (PA13)
   4. JTCK-SWCLK (PA14)
   5. NRST

Note: STLink or STM32F446 Nucleo-64 programmer board can be used for programming NLN 500b module.

11 Reference Schematic

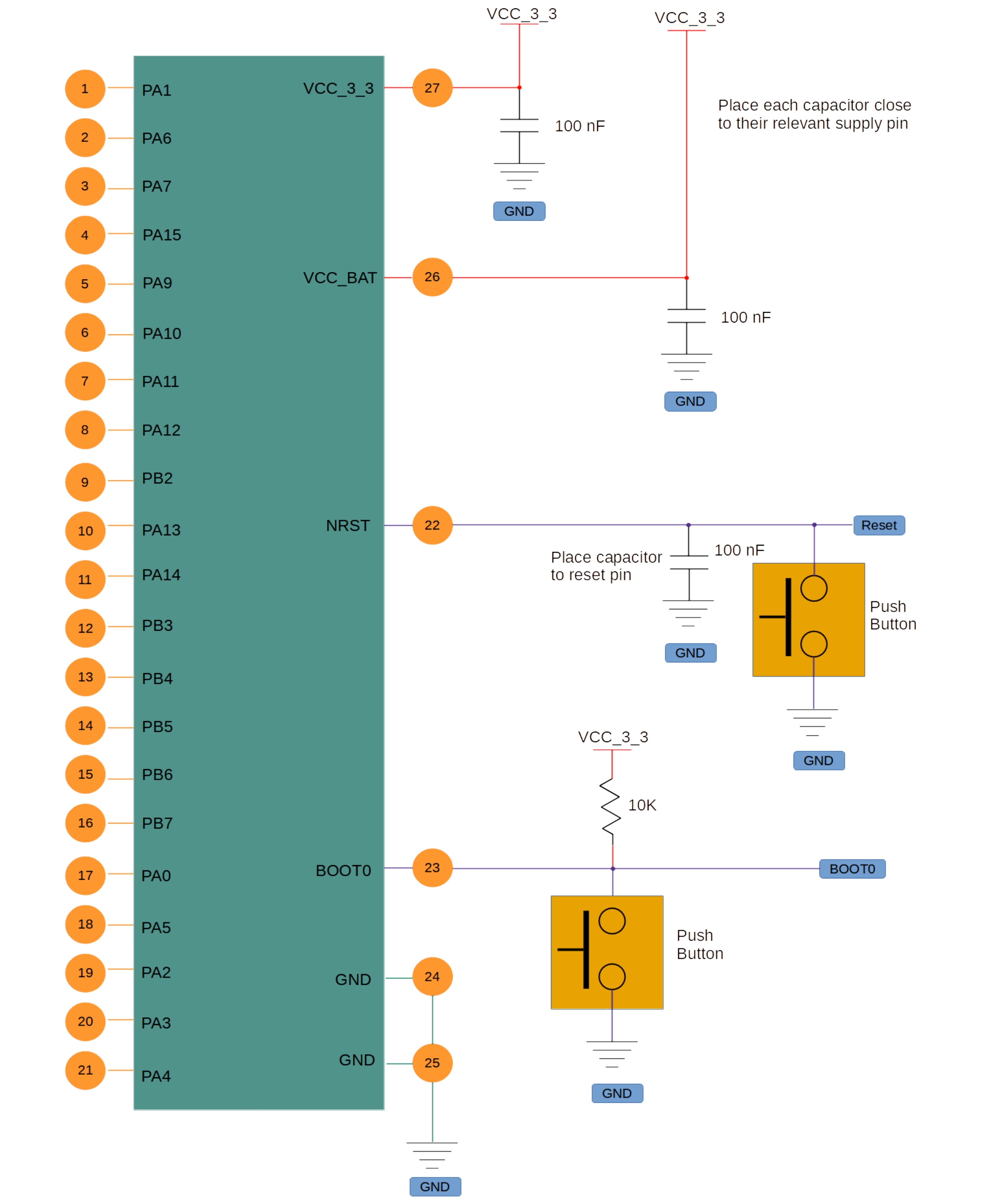

Figure 6: Reference Schematic

12 Package Outline

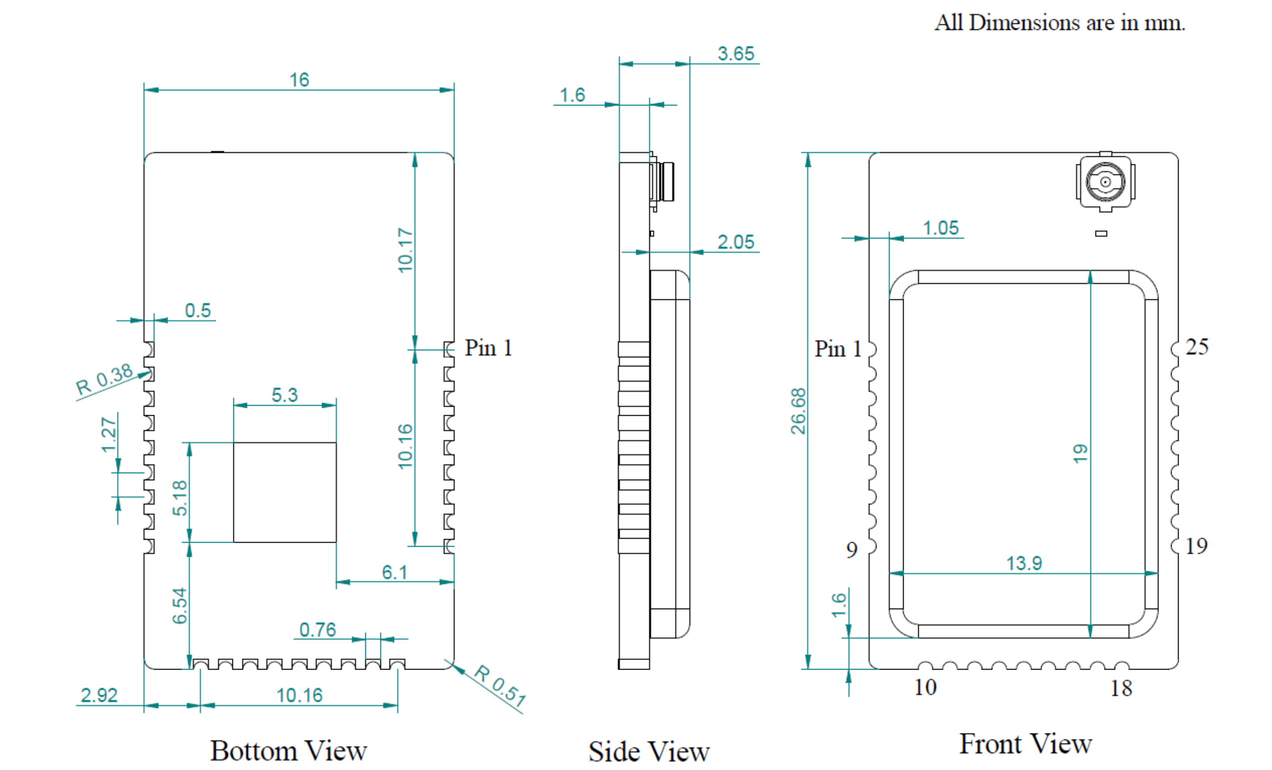

Figure 7: Size & Dimensions

13 Recommanded Footprint

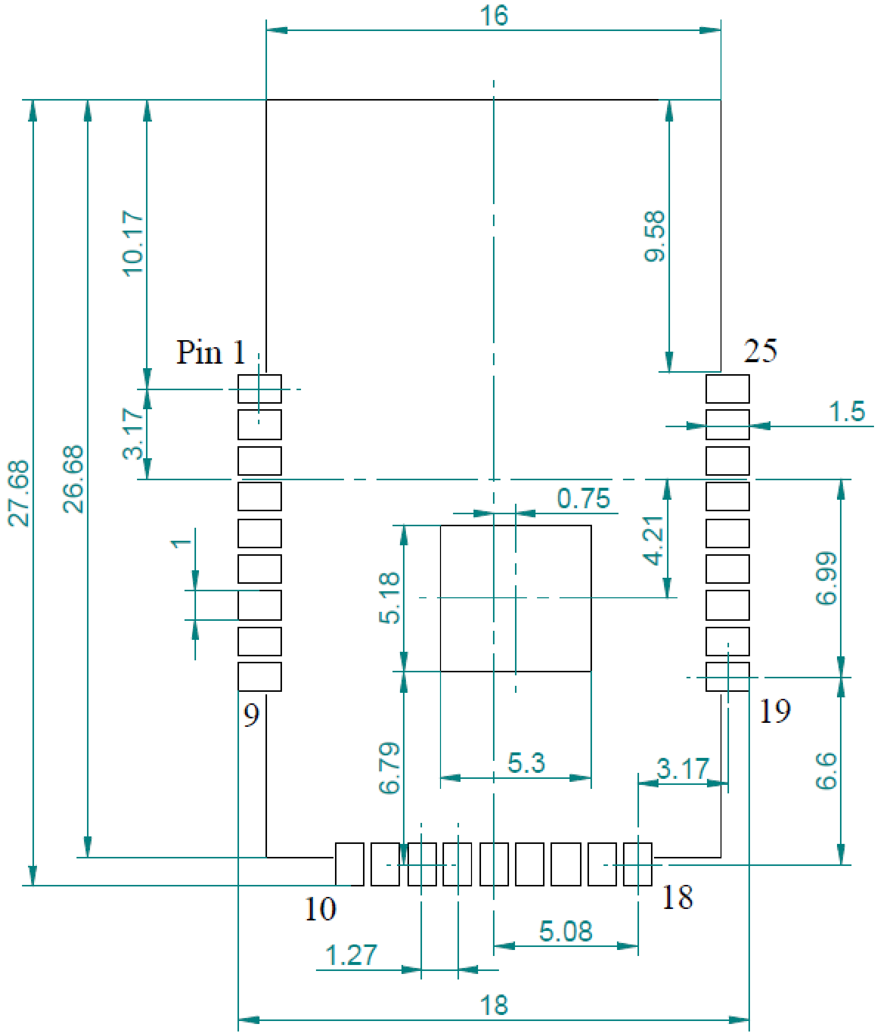

Figure 8: Recommanded Footprint

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14 Layout Guidelines

The PCB design has used enough ground area which increases the antenna efficiency. This allows stand-alone operation without any additional ground plane however care must be taken when mounting this module onto another PCB.

"The most critical aspects of radio frequency (RF) circuitry are addressed and it is highly recommended to follow these design guidelines to achieve best RF performance."

1) Keep digital and power traces/wiring away from RF circuits; clock and high-frequency circuits are the principal sources of interference and radiation, and must be placed separately and away from sensitive circuits.

2) The RF line has a 50-ohm impedance, so the line width should be as wide as feasible to match the impedance requirements.

3) Avoid parallel routing and crossing of RF traces and from under signal pads.

4) Provide good solid ground by using multiple vias.

5) The recommended practice is to use a solid (continuous) ground plane on Layer 2, assuming Layer 1 is used for the RF components and transmission lines.

6) A common practice is to use a "star" configuration for the power-supply routes.

7) Most ICs require a solid ground plane on the component layer (top or bottom of PCB) directly underneath the component. This ground plane will carry DC and RF return currents through the PCB to the assigned ground plane.

15 Placement Guidelines

The PCB design has used enough ground area which increases the antenna efficiency. This allows stand-alone operation without any additional ground plane however care must be taken when mounting this module onto another PCB.

The area around the antenna must be kept clear of any conductive or metal parts for an absolute minimum of 20 mm. Any conductive objects close to the antenna could severely disrupt the antenna pattern resulting in deep nulls and high directivity in some directions. This is true for all layers of the PCB and not just the top layer.

It is recommended that the module is mounted on the edge of the host PCB with the antenna exposed to free space. It is permitted for PCB material to be below the antenna structure of the module as long as no copper traces or planes are on the host PCB in that area.

Never place any component, planes, mounting screws, or traces in the antenna keep-out area across all layers. The actual keep-out area depends on the antenna used

Do not place the antenna close to the plastic in the industrial design. Plastic has a higher dielectric constant than air. Proximity of the plastic to the antenna results in the antenna's seeing a higher effective dielectric constant. This increases the electrical length of the antenna trace and reduces the resonant frequency.

The battery cable or any electrical interface cable must not cross the antenna trace or Antenna.

The antenna must not be covered by a metallic enclosure completely. If the product has a metallic casing or a shield, the casing must not cover the antenna. No metal is allowed in the antenna near-field.

The orientation of the antenna should be in line with the final product orientation so that the radiation is maximized in the desired direction

There must not be any ground directly below the antenna.

16 Soldering Guidelines

For re-flow soldering, it is recommended to follow the reflow profile in below figure as a guide, as well as the paste manufacturer's guidelines on peak flow temperature, soak times, time above liquid and ramp rates.

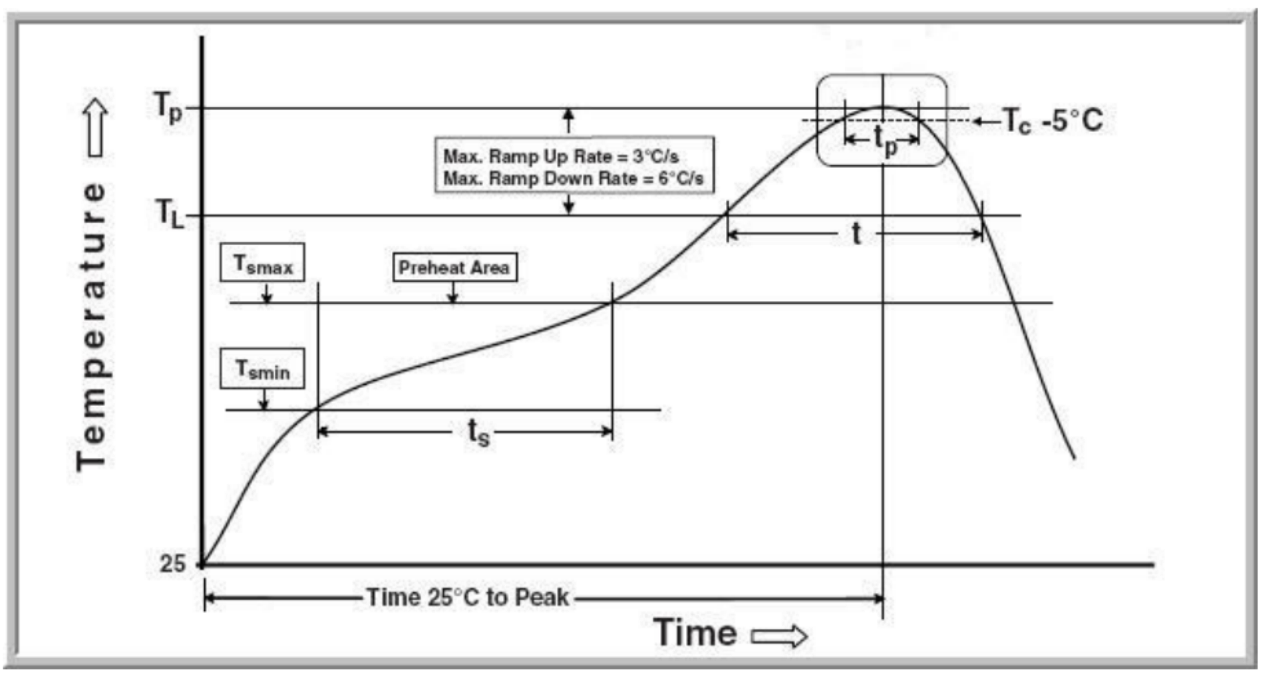

Figure 9: Reflow

Table – 10 Soldering Information

Profile Feature		Pb-Free Assembly
Average Ramp-Up Rate (TSmax to TP)		3°C / second max.
Preheat	Temperature Min (Tsmin) Temperature Max (Tsmax) Time (tSmin to tSmax)	150°C 200°C 60-180 seconds
Time maintained above	Temperature (TL) Time (tL)	217°C 60-150 seconds
Peak/Classification Temperature (TP)		245°C
Time within 5°C of the specified Peak Temperature (tP)		20-40 seconds
Ramp-Down Rate (TP to TL)		6°C / second max.
Time 25°C to Peak Temperature		8 minutes max.

17 Regulatory Approval

The module, NLN500b, is under the certification process for WPC.